Introduction

Cohen's d is the ratio of a mean difference to a standard deviation. It standardises effects for cross-study comparison and power analysis. Although simple in idea, several variants differ in which SD is used as the denominator.

Prerequisites

Means and standard deviations, two-sample t-test.

Theory

Classical definitions:

- **Cohen's** d (two independent groups): $d = (\bar{x}_1 - \bar{x}_2)/s_{\text{pooled}}$, where $s^2_{\text{pooled}} = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2]/(n_1 + n_2 - 2)$.
- **Hedges'** g: Cohen's d multiplied by a small-sample bias correction factor.
- **Glass's** Δ: d using only the control group's SD (for unequal variances).
- **Cohen's** d_z (paired): mean paired difference / SD of differences.
- **Cohen's** $d_{\mathbf{rm}}$ (repeated measures): uses a specific variance of the mean difference.

Cohen's thresholds: 0.20 / 0.50 / 0.80 small / medium / large. These are starting points; clinical context should override conventions.

Assumptions

- Approximately normal data (for the SD to be meaningful).
- Equal variances for classical Cohen's d; use Glass or Welch-style for unequal.

R Implementation

```r
library(effectsize)
set.seed(2026)

g1 <- rnorm(50, 50, 10)
g2 <- rnorm(50, 55, 10)

cohens_d(g1, g2)
hedges_g(g1, g2)
glass_delta(g1, g2)

# Paired
pre  <- rnorm(30, 50, 10)
post <- pre + rnorm(30, -5, 5)
cohens_d(pre, post, paired = TRUE)

# Unequal variances -> Glass' Delta
g1_big <- rnorm(50, 50, 5)
g2_big <- rnorm(50, 55, 15)
cohens_d(g1_big, g2_big)
glass_delta(g1_big, g2_big)
```

Output & Results

For two-sample example: Cohen's $d \approx -0.47$, Hedges' $g \approx -0.47$, Glass's $\Delta \approx -0.50$.

Interpretation

"The intervention reduced anxiety by 5.1 points (SE 2.0, Cohen's $d = 0.51$, 95 % CI 0.11 to 0.90), a medium-sized effect."

Practical Tips

- Use Hedges' g in meta-analysis; it's the small-sample-corrected version that combines across studies fairly.
- Paired designs should use d_z, not two-sample d; the two are not interchangeable.
- Report both the standardised effect and the raw effect in units that matter clinically.
- Cohen's thresholds vary across fields; cite a field-appropriate reference.
- Confidence intervals for d (e.g., from `effectsize`) are asymmetric; do not assume symmetric.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation